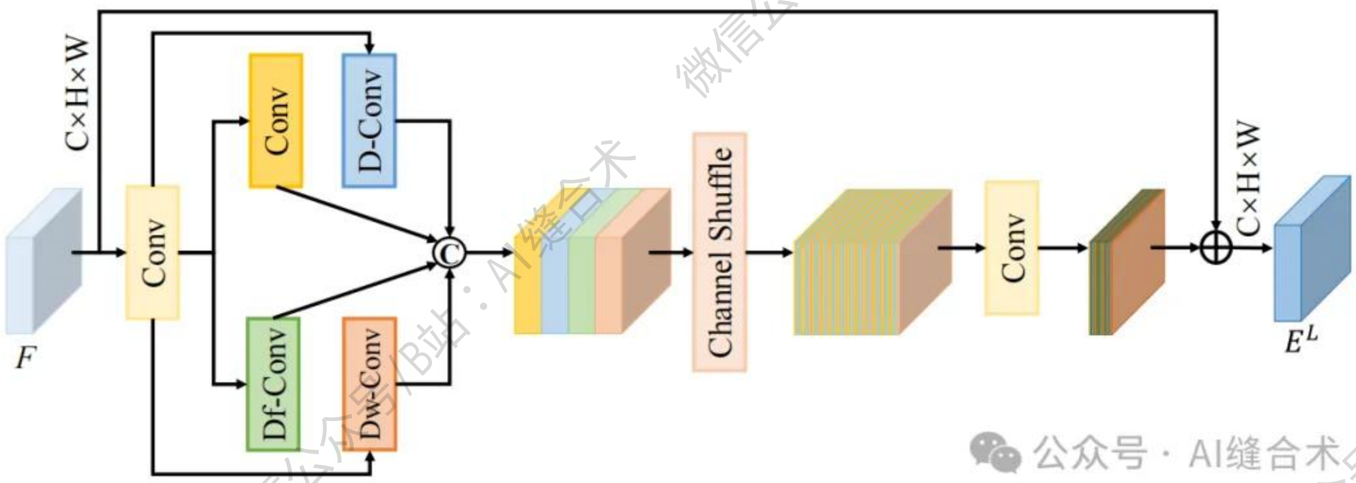
核心速览



论文题目： FreDFT: Frequency Domain Fusion Transformer for Visible-Infrared Object Detection

中文题目： FreDFT：可见-红外目标检测中的频域融合 Transformer

论文链接： https://arxiv.org/pdf/2511.10046

所属单位： 西北工业大学等

核心速览：

针对可见光-红外目标检测中模态信息不平衡及现有方法忽视频域优势的问题，本文提出频域融合 Transformer（FreDFT）。该方法通过多模态频域注意力（MFDA）挖掘模态互补信息，结合频域前馈层（FDFFL）增强特征，并设计跨模态全局建模模块（CGMM）和局部特征增强模块（LFEM）消除信息不平衡，在多个数据集上性能优于现有方法。

https://mp.weixin.qq.com/s/qmZQSSug3OW-NA-J_gNmvw

```
LFEM(
  (CBSk1): Conv(
    (conv): Conv2d(32, 32, kernel_size=(1, 1), stride=(1, 1), bias=False)
    (bn): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (act): SiLU()
  )
  (CBSk3): Conv(
    (conv): Conv2d(32, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
    (bn): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (act): SiLU()
  )
  (dconv): Sequential(
    (0): Conv2d(32, 32, kernel_size=(3, 3), stride=(1, 1), padding=(3, 3), dilation=(3, 3), bias=False)
    (1): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (2): SiLU()
  )
  (dfconv): DeformConv2d(
    (zero_padding): ZeroPad2d((1, 1, 1, 1))
    (conv): Conv2d(32, 32, kernel_size=(3, 3), stride=(3, 3), bias=False)
    (p_conv): Conv2d(32, 18, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (m_conv): Conv2d(32, 9, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
  )
  (bn): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  (silu): SiLU()
  (dwconv): Sequential(
    (0): Conv2d(32, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), groups=32, bias=False)
    (1): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (2): SiLU()
  )
  (CBS4C): Conv(
    (conv): Conv2d(128, 32, kernel_size=(1, 1), stride=(1, 1), bias=False)
    (bn): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (act): SiLU()
  )
)
input_tensor_shape  : torch.Size([2, 32, 64, 64])
output_tensor_shape : torch.Size([2, 32, 64, 64])

哔哩哔哩/微信公众号: AI缝合术, 独家整理!
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